BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 039 · Programming Techniques · Iteration and Tracing

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What prints after int x=2; x=x*3;?

Q2. What does a queue remove first?

Q3. Why should loops have a stopping condition?

Q4. What is 0 XOR 1?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Recall the term — iteration

Recall the term — accumulator

Recall the term — counter

Recall the term — condition

Recall the term — termination

New learning

Explain the important decision — total = 0; for (int i = 1; i <= 3; i++) total += i; ends with total = 6.

Worked example

Trace an accumulator

1. Identify the task — Trace an accumulator

2. Follow the logic — total = 0; for (int i = 1; i <= 3; i++) total += i; ends with total = 6.

3. Check the outcome — Explain how each step contributes to the result. Test an edge case.

We Do · Practise with guidance

Problem 1

for i = 0 to 3 exclusive, how many iterations?

Discuss, then respond

Check your thinking

3: i=0,1,2.

Problem 2

while (x < 5) x++; starting x=3: final x?

5.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Apply independently
Trace the final total for 2 + 4 + 6 using an accumulator. [3]

Question 2

Apply independently
Give a reason to choose a while loop rather than for. [2]

Question 3

Apply independently
Write a C# loop to output 1, 2, 3. [3]

Exit Ticket · Answer before checking

True or false: a while loop always runs at least once.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 6

Q2. The oldest item

Q3. To avoid non-termination

Q4. 1

You Do pages · Independent questions

Q1. 0 → 2 → 6 → 12.

Q2. When repetitions depend on a condition rather than a fixed count.

Q3. for (int i=1; i<=3; i++) Console.WriteLine(i);

Exit Ticket · Model answer

False. A while loop may execute zero times if its condition is false at the start.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: